

Arpan Pathak

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🇺🇸 Visa: Requires H1B Transfer sponsorship

📌 SUMMARY

Senior Software Engineer, **8 years: AI inference serving, backend microservices**, cloud-native infrastructure at **Amazon, Microsoft, Oracle, Razorpay**. Strong **Python, modern memory-safe C++, Rust, CUDA; BERT inference at 5,000 QPS** (ONNX, INT8) on **Docker/Kubernetes** with CI/CD; passionate about **CV/DL for road safety** on **NVIDIA Jetson Orin** (TensorRT).

📌 PERSONAL PROJECTS

- 🔗 **Driving-CivicSense: Full-Stack Edge AI (Rust, CUDA), NVIDIA Jetson Orin Nano Super NX** | [github](#)
- **Full-stack data collection:** **50,000 miles** of driving footage across all weather and terrain via vehicle camera sensors and **LIDAR**; simulation-driven synthetic data; privacy-first, on-device processing on edge neural chips.
 - **Multi-chip edge system:** **YOLOv8/v11 INT8** → **TensorRT** at ~ **12 ms** on **Jetson Orin Nano Super NX**; sensor data streamed to a **Raspberry Pi Zero 2 W UDP server**; validated on **Raspberry Pi 5** and **Orange Pi Zero**.
 - **Containerized VLM inference:** building edge VLM serving with **vLLM** and **TensorRT-LLM** in Docker containers on the Jetson.
 - **CUDA-Oxide (Rust-to-CUDA):** contributing GPU kernels and documentation fixes (unsafe intrinsics wrapped in explicit unsafe blocks); pushing Jetson limits with idiomatic, memory-safe, fearlessly concurrent kernels.
 - **Research & safety standards:** paper on **Intersection Blockage Prediction** | [paper](#); studying ISO 26262. Books: **Seeing Machines · CUDA Kernels** | [book](#)

📌 EXPERIENCE

- 📌 **Software Developer II, Microsoft, Redmond, WA** Jun 2025 - Aug 2026
- **Data Protection Services:** built and maintained **data loss prevention (DLP)** and **data lifecycle management** services with a **C#/ .NET** backend.
 - **Memory Safety (C++):** refactored a **legacy C++ codebase** with **modern C++** practices (**smart pointers, RAII**), fixing **use-after-free** and **buffer-overflow** vulnerabilities.
 - **Automation:** wrote **Python** scripts to automate internal workflows and cut manual effort.
↳ **TechStack:** **C++, C#, .NET, Python, Kubernetes, Docker, CI/CD, Linux**
- 📌 **Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA** Oct 2024 - Mar 2025
- **Backend Services & SDKs:** owned the **Terraform provider** (developer SDK) for Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month** projected savings.
 - **High-Performance Systems:** lock-free data structures reducing contention and latency in high-concurrency read-write paths.
 - **Security & Key Management:** integrated OCI Security Vault with automated key rotation across control-plane services.
↳ **TechStack:** **Java, Go, REST/gRPC, Terraform, OCI**
- 📌 **Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India** Mar 2021 - Oct 2024
- **Production Inference Serving:** scaled low-latency **BERT-based inference** to **5,000 QPS** (ONNX Runtime, INT8 quantization, latency tuning); validated throughput/latency; cut triage SLA **7 days** → **2 hours**.
 - **Real-Time ML Ranking:** owned end-to-end (training, serving, latency tuning) XGBoost + Deep Learning ensemble across **FreeVe, Twitch, MiniTV, Prime Video, Amazon Retail**; **5% ad CTR lift**, **\$5.4M/month** revenue.
 - **Event-Driven Microservices:** led a team of 3 building **programmatically guaranteed (PG) ad delivery** at **< 50 ms** via real-time CDC (Kotlin); REST/gRPC APIs.
 - **Distributed Systems:** designed a uniqueness-constraint indexing SDK (**two-phase commit, serializable isolation, 3K writes/s**).
 - **Data Platform:** built the **1-petabyte** warehouse/lake (AWS Glue, Spark) for financial reporting & pay computation.
 - **Telemetry & Automation:** real-time monitoring and automated controls cut deal over-delivery ~ **30%** and budget overspend **25%**.
↳ **TechStack:** **Python, Kotlin, PyTorch, XGBoost, ONNX Runtime, Spark, Kafka, DynamoDB, Redshift, AWS**
- 📌 **SDE, Razorpay, Bengaluru, India** Aug 2020 - Feb 2021
- **Backend Microservices:** P2P Unified Payments Gateway (Go, Protobuf/gRPC): **3M+ transactions/hour** peak, **99.99% uptime**, **150ms P99**; customers: **Zomato, Swiggy, Zerodha, Groww** (AWS, GCP).
- 📌 **Software Engineer, Mindfire, Bhubaneswar, India** Aug 2018 - Feb 2020
- Java/Spring full-stack tools; AR product-search microservice (**10K+ daily queries**).

🔧 SKILLS

↳ **Languages:** Python, C++, Rust, Go, Java, Kotlin, C, Shell/Bash
↳ **AI/ML Inference:** Deep Learning, Computer Vision, PyTorch, XGBoost, BERT, LLMs, ONNX Runtime, TensorRT (INT8), vLLM, SGLang, TensorRT-LLM, YOLOv8/v11, Edge Inference, Quantization, MLOps
🔗 **GPU & Parallel Computing:** CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming, NVIDIA Jetson Orin, HPC
🔗 **Systems Programming & Performance:** Lock-Free Data Structures, Concurrency, Memory Safety, Performance Engineering, Low-Latency Systems, Computer Architecture
🌐 **Networking:** TCP/IP, mTLS, HTTP/2/3, QUIC, eBPF, XDP, Cilium, Load Balancing
📌 **Distributed Systems & Streaming:** Consensus, Replication, Sharding, Kafka, Spark, CDC, Microservices, gRPC, REST, Protobuf
☁ **Cloud & Infra:** AWS (EC2, ECS, EKS, Lambda, Kinesis, EMR, SageMaker, CDK), Azure Kubernetes Service (AKS), Google Kubernetes Engine (GKE), Kubernetes, Docker, Helm, Terraform, Kubebuilder, Istio, CI/CD
📊 **Observability:** Prometheus, Grafana, AWS CloudWatch
📌 **Data & Storage:** DynamoDB, PostgreSQL, MySQL, Redis, Amazon Redshift, Amazon S3

🎓 EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018